(21) If $f(x)=\begin{cases} x+2, & -1<x<3 \\ 5 & x=3 \\ 8-x & x>3 \end{cases}$, then at $x=3$, $f'(x)$ is

(1) 1 (2) -1 (3) 0 (4) does not exist

(22) The derivative of $f(x)=x|x|$ at $x=-3$ is

(1) 6 (2) -6 (3) does not exist (4) 0

(23) If $f(x)=\begin{cases} 2a-x, & \text{for } -a<x<a \\ 3x-2a & \text{for } x\geq a \end{cases}$, then which one of the following is true?

(1) $f(x)$ is not differentiable at $x=a$ (2) $f(x)$ is discontinuous at $x=a$

(3) $f(x)$ is continuous for all x in $\mathbb{R}$ (4) $f(x)$ is differentiable for all $x\geq a$

(24) If $f(x)=\begin{cases} ax^2-b, & -1<x<1 \\ \dfrac{1}{|x|}, & \text{elsewhere} \end{cases}$ is differentiable at $x=1$, then

(1) $a=\frac{1}{2},\ b=\frac{-3}{2}$ (2) $a=\frac{-1}{2},\ b=\frac{3}{2}$ (3) $a=-\frac{1}{2},\ b=-\frac{3}{2}$ (4) $a=\frac{1}{2},\ b=\frac{3}{2}$

(25) The number of points in $\mathbb{R}$ in which the function $f(x)=|x-1|+|x-3|+\sin x$ is not differentiable, is

(1) 3 (2) 2 (3) 1 (4) 4

SUMMARY

In this chapter we have acquired the knowledge of

- Derivative as a rate of change. If $y=f(x)$ then the derivative of y with respect to x at x_0 is $\lim\limits_{\Delta x\to 0}\dfrac{f(x_0+\Delta x)-f(x_0)}{\Delta x}$, provided the limit exists, where existence means
$$f'(x_0^+)=\lim_{\Delta x\to 0^+}\frac{f(x_0+\Delta x)-f(x_0)}{\Delta x}=\lim_{\Delta x\to 0^-}\frac{f(x_0+\Delta x)-f(x_0)}{\Delta x}=f'(x_0^-)$$
is a unique real number.
- Derivative of $y=f(x)$ at $x=x_0$ is $\left(\dfrac{dy}{dx}\right)_{x=x_0}=f'(x_0)=\lim\limits_{x\to x_0}\dfrac{f(x)-f(x_0)}{x-x_0}$.
- Geometrical meaning of the derivative of $y=f(x)$ is the slope of the tangent to the curve $y=f(x)$ at $(x, f(x))$.
- Physical meaning of the derivative of $s=f(t)$ with respect to t is the rate of change of displacement, that is velocity. The second derivative is acceleration and the third derivative is jerk.
- Discontinuity of $y=f(x)$ at $x=x_0$ implies non-differentiability of $f(x)$ at $x=x_0$.

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